

Leakage-restricted baselines under three information budgets

LEAKAGE BARRIER: target lithofacies labels y_t are hidden until final scoring

Labelled source wells W_s

curves x_s + facies y_s + confidence c_s

Unlabelled target well(s) W_t

curves x_t + measured depth d_t ; no y_t

Select and disclose the admissible information budget

1 Inductive DG

Global-LGBM

Global curves + missingness masks
source-fitted statistics only

**NO TARGET
AGGREGATES**

2 Post-drilling

Hybrid-full · target normalisation

Global + whole-well z-scores
optional relative depth

**FULL TARGET
INTERVAL**

3 While-drilling

Hybrid-while · causal normalisation

Global + causal-prefix z-scores
no total depth / relative depth

**PAST + CURRENT
ONLY**

Shared CPU LightGBM backbone

main-text feature variants | early stopping

Optional post-processing

centred depth smoothing — OFFLINE only

Unequal-budget references (not shown): Global-RF and appendix CORAL-WLP

Evaluation: P1 random-well | P2 quadrant hold-out | P3 LOWO

Macro-F1 + well-level tail risk (worst-well / worst-10%)